Fie $\Sigma \subseteq E$ și $\alpha, \beta, \gamma, \delta \in E$, a.î.

$$\Sigma \cup \{\alpha, \beta\} \vdash \gamma \to \delta$$

Dem. că $\Sigma \cup \{\neg(\delta \to \neg(\alpha \wedge \beta \wedge \gamma))\}$ e nesatisf. (1) – (2) –

Rez.:

$$\Sigma \cup \{\alpha, \beta\} \vdash \gamma \to \delta.$$

$$\Updownarrow (TD)$$

$$\Sigma \cup \{\alpha\} \vdash \beta \to (\gamma \to \delta)$$

$$\Updownarrow (TD)$$

$\Sigma \vdash \alpha \to [\beta \to (\gamma \to \delta)]$

$\Updownarrow$ (†)

$\Sigma \models \alpha \to [\beta \to (\gamma \to \delta)]$ (*)

$\varphi :\equiv \neg(\neg\delta \to \neg(\alpha \wedge \beta \wedge \gamma))$ $\in E$

$\Sigma \cup \{\varphi\} \to$ unsatisf

$\Updownarrow$

$(\nexists e : V \to \mathcal{L}_2\, (e \models \Sigma \cup \{\varphi\}))$

$$(\forall h : V \xrightarrow{\nexists} \mathcal{L}^2)\,(h \neq \Sigma \cup \{\varphi\})$$

$$(\forall h : V \xrightarrow{\nexists} \mathcal{L}^2)\,(h \neq \Sigma \text{ deu } h \neq \varphi)$$

$$(\forall h : V \xrightarrow{\nexists} \mathcal{L}^2)\,(h \neq \Sigma \Rightarrow h \neq \varphi).$$

#

Fix $l_i : V \to \mathcal{L}_n$ a.r. ~~lift~~

$l_i \models \Sigma$.

$\Downarrow (*)$

$1 = \widetilde{l_i}(\alpha \to \neg\beta \to (\gamma \to \delta))$

$= \widetilde{l_i}(\alpha) \to \neg\widetilde{l_i}(\beta) \to (\widetilde{l_i}(\gamma) \to \delta)$

$= \overline{\widetilde{l_i}(\alpha)} \vee \overline{\neg\widetilde{l_i}(\beta)} \vee \overline{\widetilde{l_i}(\gamma)} \vee \widetilde{l_i}(\delta)$

$(*)$

Bsp. abs. Int. φ: II

$$\tilde{I}_n(\varphi) = 1$$

$$\| $$

$$\tilde{I}_n(\neg\delta \to \neg(\alpha \wedge \beta \wedge \gamma))$$

$$= \overline{\tilde{I}_n(\delta)} \to \overline{\tilde{I}_n(\alpha) \wedge \tilde{I}_n(\beta) \wedge \tilde{I}_n(\gamma)}$$

$$= \overline{\overline{\tilde{I}_n(\delta)}} \vee \overline{\tilde{I}_n(\alpha) \wedge \tilde{I}_n(\beta) \wedge \tilde{I}_n(\gamma)}$$

$$= \overline{\tilde{I}_n(\alpha)} \vee \overline{\tilde{I}_n(\beta)} \vee \overline{\tilde{I}_n(\gamma)} \vee \tilde{I}_n(\delta)$$

$$\Downarrow$$

$$\overline{\ln(\alpha) \vee \ln(\beta) \vee \ln(\gamma)} \vee \ln(\delta) = \overline{1} = 0$$

or (**).